Decision Theory: curriculum and assessment

In this course you describe a decision problem as a Markov Decision Process (MDP): a model with states, actions, rewards and probabilities. A policy is a rule that says which action to take in each state. When you know the model, you can calculate a good policy from it. When you do not know the model, you can learn a good policy from data, by trying actions and seeing what happens. The methods that do this are known in AI as reinforcement learning (RL). You apply all of it to an MDP of your own.

LECTURER	CLASS	MIDTERM
Iman Mossavat	**Thursday afternoon, 9 sessions**	**Mon 30 Nov, afternoon**

EXAM
Mon 18 Jan, afternoon, on paper

The course in one line

Your project grows next to the theory. We meet on Thursday afternoons. Week 13 has a small midterm on Monday afternoon. The last two weeks are for quizzes, your project and the exam.

9	10	11	12	13	14	15	·	16	17	18
S1	S2	S3	S4	mid	S5	S6	break	S7	S8	S9

Session · Midterm · Christmas break, 21 Dec – 3 Jan · Quizzes, review, presentations · Exam

Week by week

Every session has the same shape: a short quiz on last week, a lecture block, then hands-on work on your own MDP. The homework of each week is small and feeds the next quiz.

SEMESTER WK **9**
Session 1
Thu 5 Nov

Your first MDP

IN CLASS	Workshop: design a small MDP on paper, simulate it, present it. No solver yet.
READ	Sutton & Barto 3.1–3.3
HOMEWORK	Finish the result sheet for your MDP (uploaded once, completion only).

SEMESTER WK **10**
Session 2
Thu 12 Nov

Returns, policies, V and Q

IN CLASS	Reading a Q-table. Spotting a flawed MDP. When is RL a sensible tool?
READ / WATCH	S&B 3.1–3.4 · video part 1
QUIZ 1	MDP design, the Markov property, return.
HOMEWORK	Write the V equations for a fixed policy on a 2–3 state slice of your MDP.

SEMESTER WK **11**
Session 3
Thu 19 Nov

Bellman equations for a fixed policy SPRINT 1 ENDS

IN CLASS	Solve small systems by hand. Iterative policy evaluation in the course toolkit.
READ / WATCH	S&B 3.5–3.8 · video part 2 (first half)
QUIZ 2	Returns, the meaning of V and Q.
HOMEWORK	Code policy evaluation for your MDP.

SEMESTER WK **12**
Session 4
Thu 26 Nov

Optimal policies: value iteration and policy iteration

IN CLASS	One value-iteration sweep by hand. One policy-improvement step. Generalized policy iteration.
READ / WATCH	S&B ch. 4 · video part 2 (second half)
QUIZ 3	Bellman equations for a fixed policy.
HOMEWORK	Run value iteration on your MDP and compare with the expert Q-table.

SEMESTER WK **13**
Midterm
Mon 30 Nov

Midterm week SMALL MIDTERM

Use this week to finish your MDP solved with the model. If it does not work, show where it fails, with a clear analysis of the failure. There is no Thursday session this week. The small midterm is on Monday 30 November, in the afternoon. `format to be confirmed`

SEMESTER WK **14**
Session 5
Thu 10 Dec

Monte Carlo SPRINT 2 ENDS

IN CLASS	Estimate V from episodes. How many episodes do you need? Confidence intervals, back to statistics.
READ / WATCH	S&B ch. 5 · video part 3
QUIZ 4	Value iteration and policy improvement.
HOMEWORK	Monte Carlo estimate with a confidence interval for your MDP.

SEMESTER WK **15**
Session 6
Thu 17 Dec

Exploration and off-policy learning SPRINT 3 ENDS

IN CLASS	ε-greedy exploration, learning a policy from episodes, learning about one policy while following another.
READ / WATCH	S&B ch. 5 (rest) · video part 3 (rest)
QUIZ 5	Monte Carlo and confidence intervals.
HOMEWORK	ε-greedy control on your MDP, averaged over several runs.

NO SESSIONS
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21 Dec – 3 Jan

Christmas break

Sprint 3 is delivered before the break. After the break we start with Q-learning. If you like, start on video part 4 and S&B ch. 6 during the break. optional

SEMESTER WK **16**
Session 7
Thu 7 Jan

Temporal-difference learning and Q-learning

IN CLASS	One Q-learning update by hand. Off-policy learning. Q-learning compared with value iteration.
READ / WATCH	S&B ch. 6 · video part 4
QUIZ 6	Exploration, ε-greedy, off-policy learning.
HOMEWORK	Q-learning on your MDP. Compare its policy with the expert's.

SEMESTER WK **17**
Session 8
Thu 14 Jan

Review, quiz and project presentations SPRINT 4 ENDS

IN CLASS	A cumulative quiz on everything so far, practice for the exam, and your project presentation.
QUIZ 7	Temporal-difference learning and Q-learning, plus a mix of earlier topics.
HOMEWORK	Prepare your cheat sheet for the exam.

SEMESTER WK **18**
Exam
Mon 18 Jan

Exam WRITTEN EXAM

MONDAY AFTERNOON	Written exam on paper, 90 minutes plus 30 extra.
THURSDAY 21 JAN	Last session: wrap-up and last project work.

How you are assessed

All graded work on theory is done in class, on paper, without AI. For code you may use an AI assistant, as long as you understand every mechanism in it. I do not grade your code. I grade what you understand.

The week of the 1-1 discussion about how you built your project will be announced. `to be confirmed`

Mon 18 Jan

Written exam

- On paper, in class, 90 + 30 minutes.
- Same style as the quizzes: small MDP from a story, Bellman equations, one Monte Carlo or Q-learning update by hand.
- Marks are for reasoning. A correct answer without working earns nothing.

Weeks 10–17

Quizzes

- Seven short quizzes at the start of sessions 2–8, on last week's topic.
- They are practice for the exam, in the same style, and they give you feedback on where you stand.
- One question is about your own homework or MDP.
- I mark them within the week. Going through them is part of the next session.

Weeks 9–17

Project: your own MDP

- One MDP that you keep all course. You deliver it in four sprints, each with its own target.
- It is a team project, normally with two students. Working alone is allowed.
- In a team, each student's personal contribution must be clear, and both students must have a clear technical contribution.
- You present it in week 17 (Thursday 14 January).
- Graded on your understanding as you show it, not on the code.

Required, not graded

Homework

- About 1–2 hours a week. Ask me for feedback whenever you want it.
- Give me access to your portfolio and submit your homework there.
- It can appear in a quiz.
- You must submit all of it to pass the semester.

Project sprints

You deliver the project in four sprints. Each sprint has one target. In a team, show in every sprint what each student contributed. Sprint 3 is due before the Christmas break.

SPRINT	ENDS	TARGET
1	Week 11 Thu 19 Nov	**Your MDP, tested.** Idea and story agreed with me. Your MDP code tested on a tiny MDP (2 states, 2 actions) whose expected behaviour you can argue for.
2	Week 14 Thu 10 Dec	**Solved with the model.** You explain your model and code. Policy evaluation and value iteration on your MDP, checked against the expert Q-table.
3	Week 15 Thu 17 Dec	**Learned from episodes.** Monte Carlo and ε-greedy on your MDP, compared with the answer from the model. Tests that show you can trust your results. A minimum viable answer to every item.
4	Week 17 Thu 14 Jan	**Q-learning, final delivery and presentation.** Q-learning on your MDP, compared with the model and with Monte Carlo. What you changed after the feedback, your conclusions, and your presentation.

To pass

- A grade of 5.5 or higher on the exam and on the midterm.
- An acceptable end project, delivered gradually in sprints.
- A 1-1 discussion with me about how you built your project.
- All homework submitted.

Materials

Book

Sutton & Barto, *Reinforcement Learning: An Introduction*, 2nd edition. Free from the authors.

Download the PDF

Slides and course files

The slides of the course, a workshop reader and a Python notebook. The slides are the 2023–24 version and will be updated during the course.

Open the course folder

Video series

These videos are not by Sutton and Barto. They are by the YouTube channel **Mutual Information**, in the series *Reinforcement Learning, by the Book*. They follow the chapters of the book. Watch parts 1–4. Part 5 is not part of this course.

- Part 1 · S&B 3.1–3.4
- Part 2 · S&B 3.5–3.8, ch. 4
- Part 3 · S&B ch. 5
- Part 4 · S&B ch. 6

On Canvas

The course toolkit (Python with numpy and matplotlib) and the MDP idea sheets. You need a laptop with Python.

Draft for discussion. Weeks can still change.